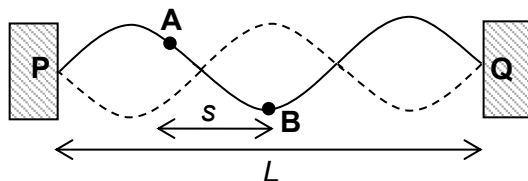


- 17 A guitar string of length L is stretched between two fixed points **P** and **Q** and made to vibrate transversely as shown below.



Two points **A** and **B** on the string are separated by a distance s . The maximum kinetic energies of points **A** and **B** are K_A and K_B respectively. Which of the following gives the correct phase difference and relationship between maximum kinetic energies of the points?

		Phase difference	Maximum kinetic energy	
	A	$\frac{3s}{2L} \times 360^\circ$	$K_A < K_B$	
	B	$\frac{3s}{2L} \times 360^\circ$	same	
	C	180°	$K_A < K_B$	
	D	180°	same	

Ans: C

The adjacent segments of a stationary wave on a string are in antiphase. Point B has a larger amplitude of vibration than point A. Thus the maximum kinetic energy that point B can have during the vibration is greater than A.

- 18 The images of two sources are just resolved. Which of the following is a correct statement of